

David A. Williams

Controls & Automation Engineer

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PROFESSIONAL SUMMARY

Strategic Controls & Automation Engineer with 5+ years of experience in high-volume industrial environments. Expert in PLC/HMI programming (Siemens, Mitsubishi, Yaskawa) and the design of robust automated systems. Proven track record in troubleshooting complex vision systems and optimizing Ethernet/IP network architectures to maximize system uptime. Specialized in bridging the gap between mechanical design and advanced electrical control systems.

WORK EXPERIENCE

Wegmann Automotive - Murfreesboro, TN Process Engineer

Nov 2023 - Present

- Interpreted and applied electrical and mechanical design drawings to develop solutions.
- Designed and deployed a PLC system migration to Ethernet/IP Remote I/O, increasing data acquisition and reducing network latency across the production line.
- Designed and commissioned custom automated control systems that boosted production throughput, utilizing Siemens TIA Portal and Ignition SCADA.

Wegmann Automotive - Murfreesboro, TN Steel Assembly Technician

Nov 2021 – Nov 2023

- Optimized Cognex In-Sight vision inspection algorithms to achieve a high detection rate, ensuring zero-defect delivery to downstream processes.
- Collected and analyzed machine data to minimize scrap and downtime.
- Participated in continuous improvement initiatives in order to increase platform performance.

General Mills - Murfreesboro, TN Machine Operator

May 2019 – Nov 2021

- Identified and resolved mechanical and electrical equipment failures in high-speed food processing environments in order to maintain efficient OEE rating.
- Performed preventative and breakdown maintenance.
- Learned industry standards: KPI's, Project Management, Lean Manufacturing, and 5s.

EDUCATION

B.S Mechatronics Engineering | Middle Tennessee State University | Expected May 2026

A.A.S Mechatronics Technology | Motlow State Community College | May 2021

TECHNICAL SKILLS

- **PLCs/HMIs:** Siemens (Level 1 & 2), Mitsubishi, Yaskawa, Allen-Bradley/Rockwell.
- **Robotics/Vision:** Fanuc, Mitsubishi, Cognex In-Sight, Keyence.
- **Vision Systems:** Proficient with Cognex In-Sight software and brief experience with Keyence systems.
- **Networking/Protocols:** Ethernet/IP, Modbus/TCP, Profibus, BACnet
- **Software:** AutoCAD, SolidWorks, Fusion 360, Python, C++, Linux, Windows.